

Titanic Dataset - Exploratory Data Analysis (EDA)

Objective: To perform Exploratory Data Analysis (EDA) on the Titanic Dataset and identify patterns, trends, relationships, and anomalies using Python, Pandas, Matplotlib, and Seaborn.

Dataset Information

Dataset: Titanic Dataset

- Number of Rows: 891
- Number of Columns: 12

The dataset contains passenger information such as age, gender, ticket fare, passenger class, and survival status.

Missing Values Analysis

The Age, Cabin, and Embarked columns contain missing values.

The Cabin column has the highest number of missing values.

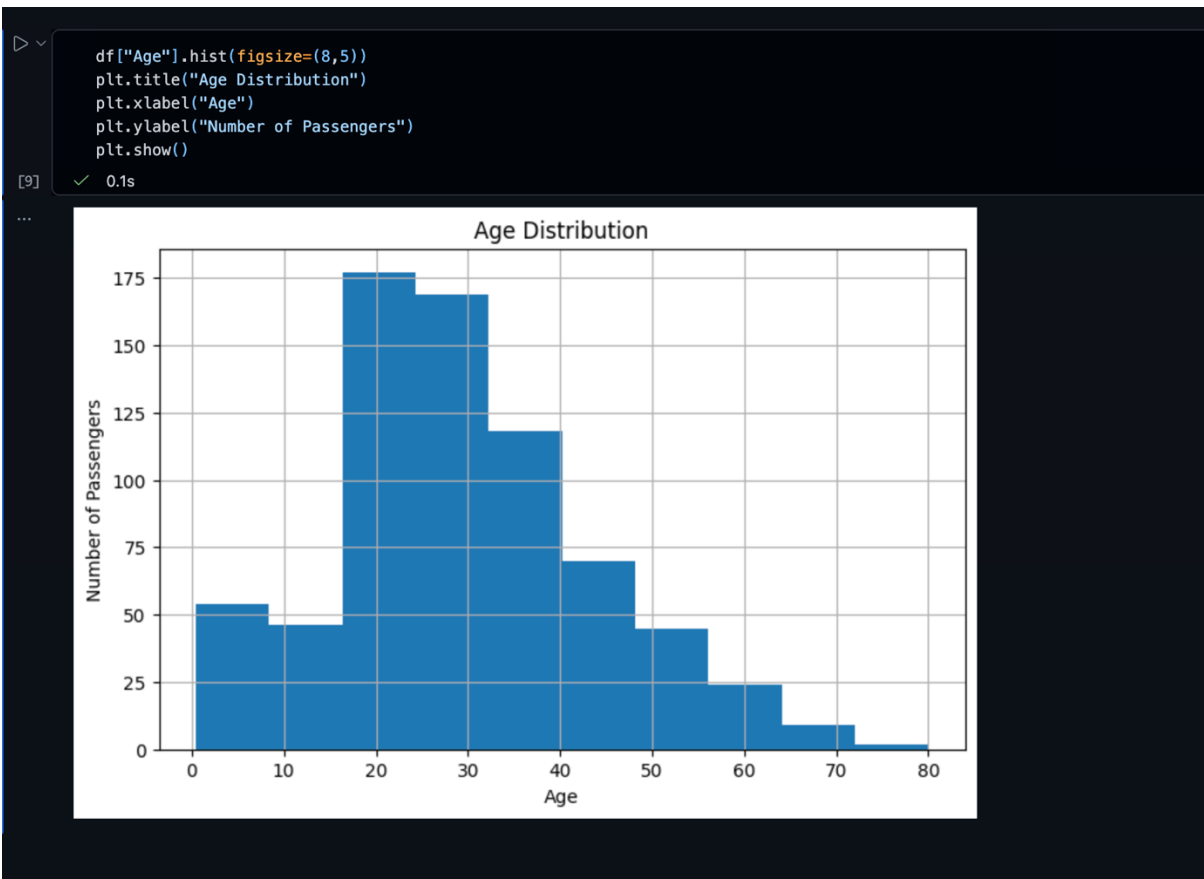


Figure 1: Age Distribution

Observation:

Most passengers were between 20 and 40 years old.

```
sns.boxplot(x=df["Fare"])  
plt.title("Fare Distribution")  
plt.show()
```

✓ 0.0s

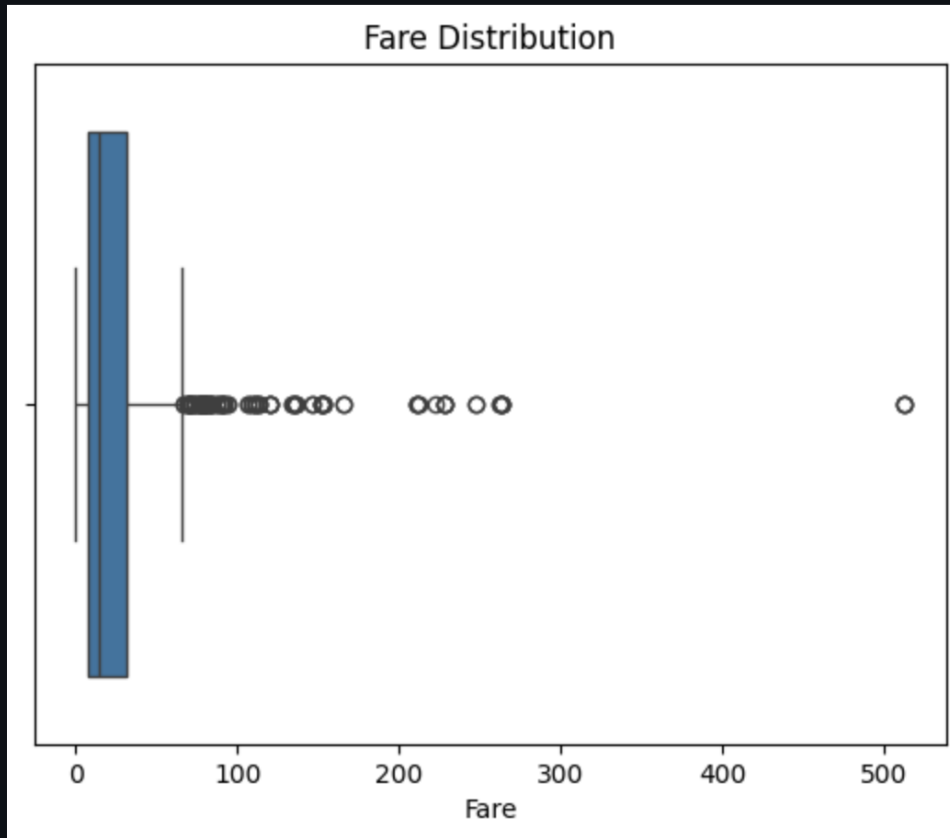


Figure 2: Fare Distribution

Observation:

Several outliers are present in the Fare column.

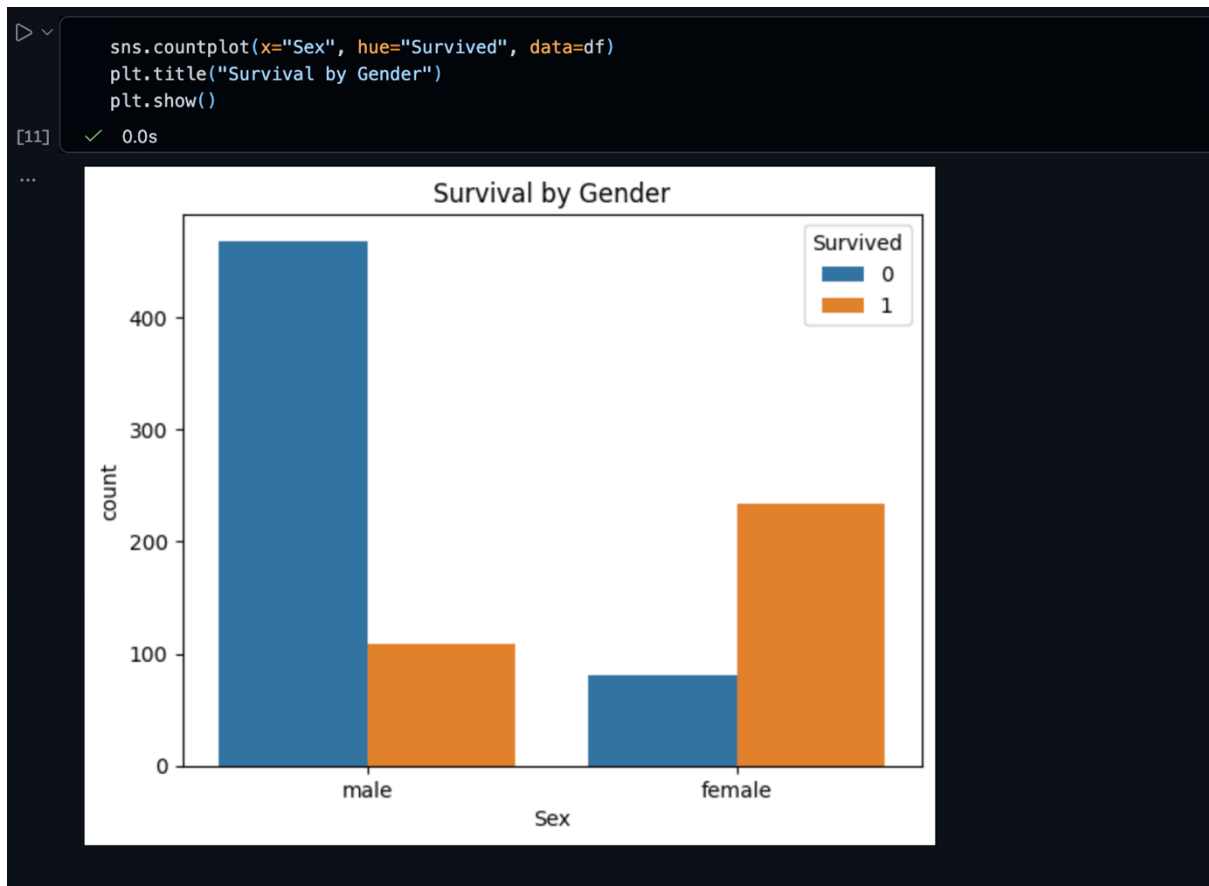


Figure 3: Survival by Gender

Observation:

Female passengers survived more frequently than male passengers.

```
sns.countplot(x="Pclass", hue="Survived", data=df)
plt.title("Survival by Passenger Class")
plt.show()
```

✓ 0.0s

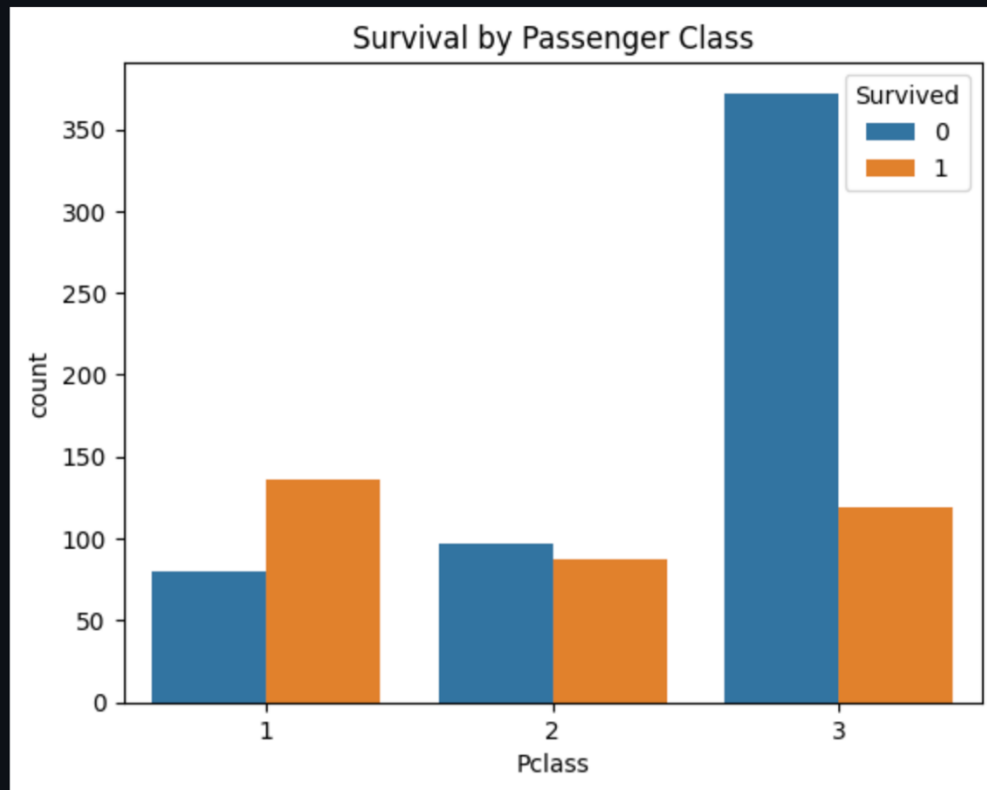


Figure 4: Survival by Passenger Class

Observation:

First-class passengers had higher survival rates than other classes.

```

numeric_df = df.select_dtypes(include=np.number)

plt.figure(figsize=(8,6))
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()

```

✓ 0.0s

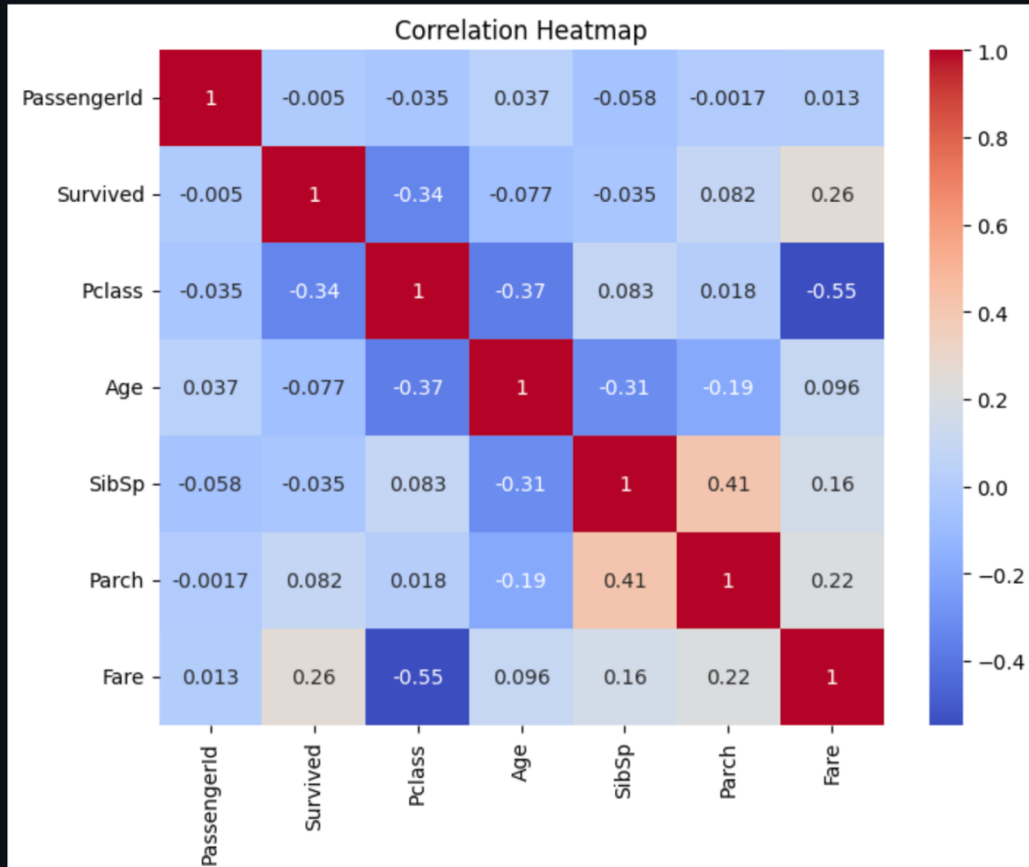


Figure 5: Correlation Heatmap

Observation:

Fare and Passenger Class show a noticeable relationship.

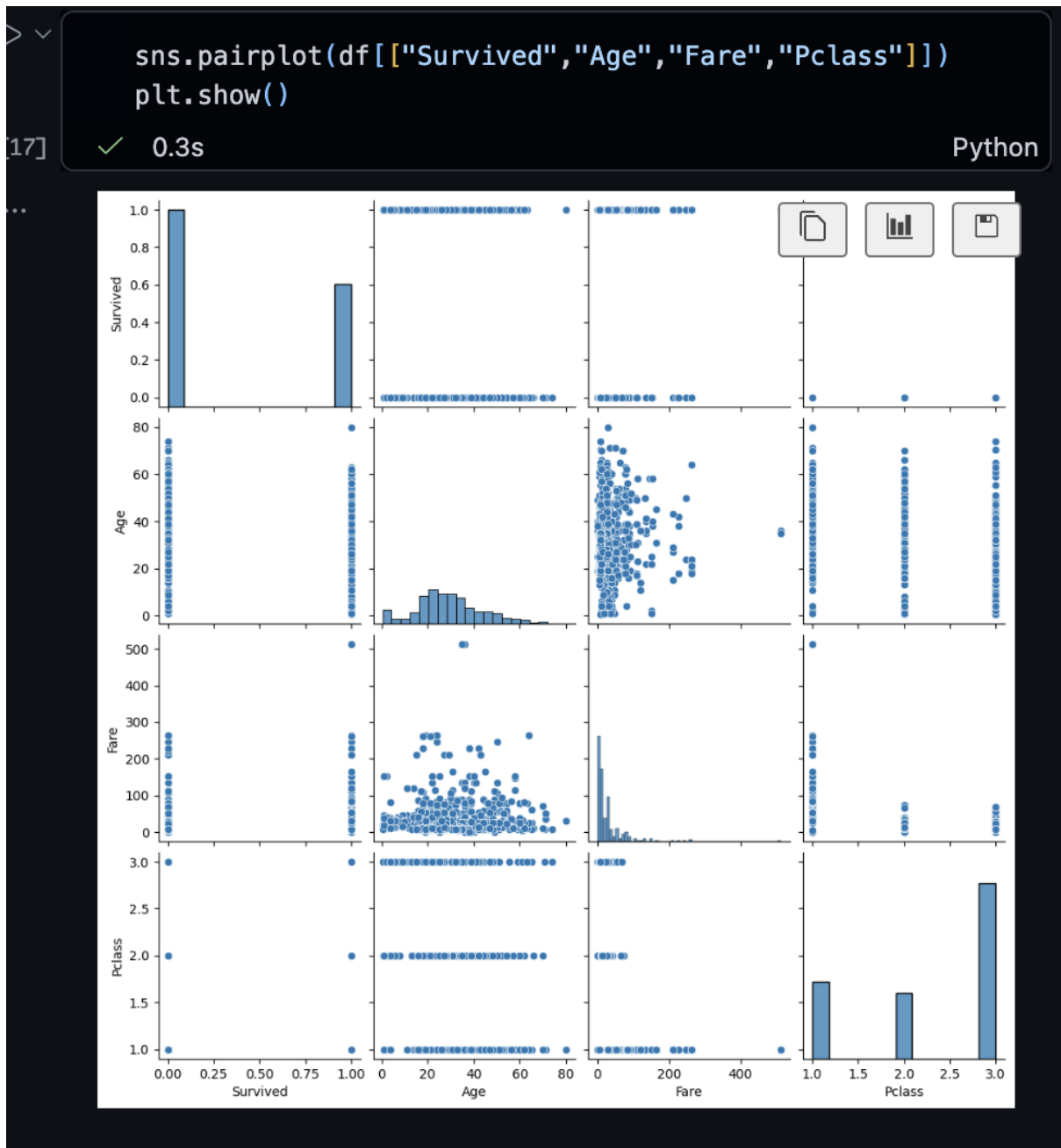


Figure 6: Pairplot

Observation:

Higher fares are generally associated with higher passenger classes.

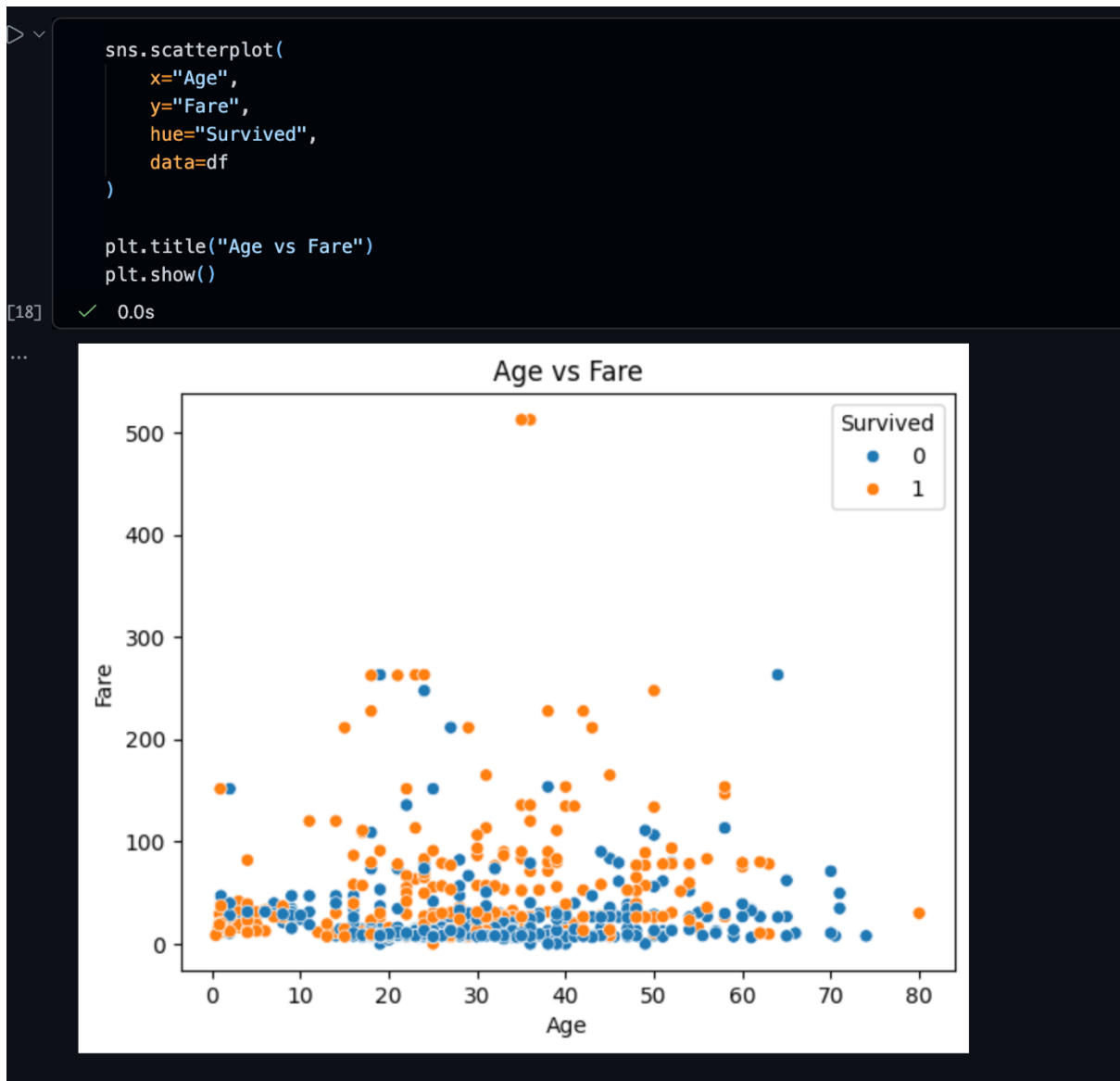


Figure 7: Age vs Fare Scatter Plot

Observation:

Passengers paying higher fares tended to have better survival chances.

Summary of Findings

1. Most passengers were between 20 and 40 years old.
2. Fare contains several outliers.
3. Female passengers survived more often than males.
4. First-class passengers had the highest survival rate.
5. Higher ticket fares were associated with better survival chances.
6. Missing values were present in Age, Cabin, and Embarked columns.
7. Fare and Passenger Class showed a significant relationship.

Conclusion

Exploratory Data Analysis helped identify important trends and survival patterns in the Titanic dataset. Passenger class, gender, and fare were important factors affecting survival.